SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2_AT3 — CASE STUDY

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO2 – Evaluation (BL5)	**Assessment:**	Assessment Tool 3 – Case Study
Weightage:	20% of CIA	**Total Marks:**	20 (2 Questions × 10 Marks)
CO2 Statement:	*Analyze real-world data scenarios and evaluate/justify the most appropriate chart types, coordinate systems, color scales, and dashboard designs to represent them.(BL5)*		
Student Name:	Tejaswini P	**Reg. No.:**	192324046
Section / Batch:	2023/B tech AIDS	**Date:**	1/8/2026

Code of Conduct

I Tejaswini P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tejaswini

Instructions

- This test contains 2 case-study questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each case study has four sub-parts (a–d); answer all parts for full marks.
- Support your answers with brief calculations or justifications where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – CASE STUDY

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT3 — Case Study

CO2 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO2-AT3 Case Study questions, in which students analyze a realistic multi-part scenario and select, justify, design, and critique appropriate visualizations across chart types, coordinate systems, color scales, and dashboard development.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of the full range of chart types, coordinate systems, and color scales relevant to the case, and correctly identifies which apply to each sub-part.	Good understanding of the relevant concepts, with only minor gaps in identifying appropriate chart types, scales, or color choices.	Basic understanding, but with some misconceptions about which visualization techniques suit the scenario's data.	Poor understanding of core concepts; proposes techniques fundamentally unsuited to the case study's data.
Explanation	**25%**	Provides detailed, well-reasoned justification for every design decision and critique, explicitly tied to specific details of the case.	Provides clear justification for most decisions and critiques, with adequate reasoning.	Justification is limited or generic; decisions are stated without sufficient reasoning tied to the case.	Little or no justification provided; answers are unsupported assertions.
Application	**20%**	Effectively applies visualization concepts to analyze and solve every sub-part of the case, including any calculations, with	Applies concepts correctly to most sub-parts, with only minor errors.	Limited application; some sub-parts are weakly addressed or only partially correct.	Fails to apply appropriate concepts; sub-parts are left unaddressed or answered incorrectly.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		well-reasoned conclusions.			
Organization	**15%**	The response is clearly structured, addressing every sub-part (a–d) in a logical sequence with coherent overall flow.	The response is organized and addresses each sub-part, with only minor issues in flow or clarity.	Basic structure; sub-parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and sub-parts are left unaddressed or answered out of order.
Accuracy	**10%**	All parts of the answer — chart choices, calculations, critiques, and dashboard design — are completely accurate and well-suited to the case.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the case, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

4) (a) Which chart type shows win/loss composition per team most clearly?
Answer:
A Stacked Bar Chart is the most suitable chart type.
Justification:

- It clearly shows both wins and losses for each team in a single bar.
- It makes it easy to compare the composition across multiple teams.
- It uses a Cartesian coordinate system, which is ideal for comparing quantities.

(b) Which chart type and coordinate system is best suited to comparing the top team's five attributes at once?
Answer:
A Radar Chart (Spider Chart) using a Polar Coordinate System is the best choice.
Justification:

- It displays multiple attributes (e.g., Attack, Defense, Speed, Passing, Stamina) simultaneously.
- The overall shape makes strengths and weaknesses easy to identify.
- Polar coordinates are well suited for multivariate comparisons of a single item.

(c) Design a dashboard combining the teams' points ranking with the top team's attribute profile, including a one-sentence narrative takeaway.
Dashboard Layout:

- Left Panel: Bar Chart showing Team vs Points (ranked highest to lowest).
- Right Panel: Radar Chart showing the Top Team's Five Attributes.
- Dashboard Title: *Team Performance Dashboard*

Narrative Takeaway:
The top-ranked team leads the league in points and demonstrates consistently strong performance across all five key attributes, contributing to its overall success.

(d) If you needed to compare all 10 teams' attribute profiles at once (not just the top team), would your chart from part (b) still work well? Explain the limitation and propose an alternative.
Answer:
No.
Limitation:

- A radar chart becomes cluttered when displaying many teams.
- Overlapping lines make it difficult to distinguish individual teams.
- Comparing attribute values accurately becomes challenging.

9) (a) What axis scale is required for the defect rate data given its range, and why?
Answer:
A logarithmic (log) axis should be used for the defect rate data because the values span a wide range. A logarithmic scale compresses larger values while preserving smaller ones, making it easier to compare suppliers with both low and high defect rates. A linear axis would make smaller defect rates appear almost indistinguishable.

(b) Design a chart combining supplier shipment volume and on-time delivery rate, annotating the supplier posing the greatest risk.
Recommended Chart:

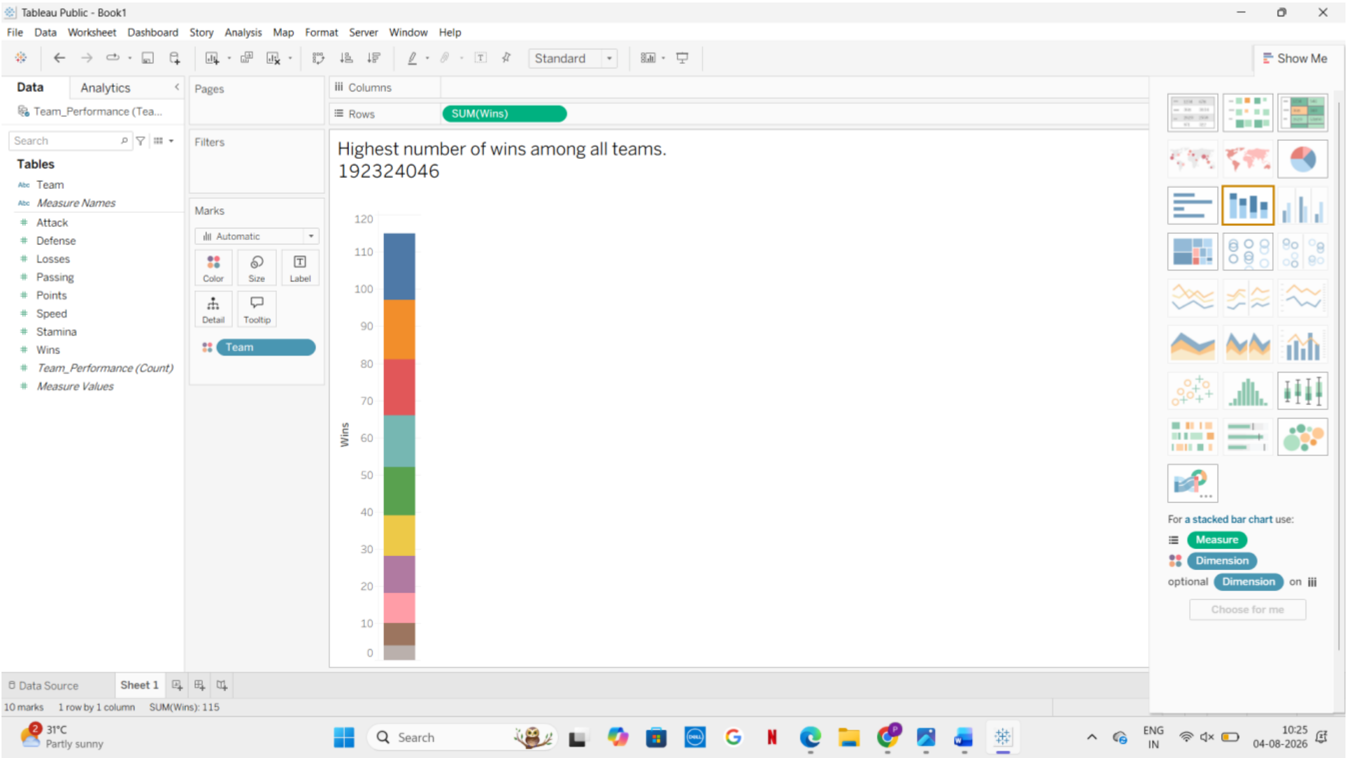
- Bar Chart: Shipment Volume
- Line Chart: On-Time Delivery Rate
- Coordinate System: Cartesian Coordinate System
- Chart Type: Dual-Axis Combination Chart

Annotation:
Supplier X has high shipment volume but a low on-time delivery rate, making it the highest operational risk and requiring immediate attention.

(c) Propose a chart for the 2-year production volume trend, and explain how you would highlight any seasonal patterns.
Recommended Chart:
A Line Chart with Month on the X-axis and Production Volume on the Y-axis.
How to Highlight Seasonal Patterns:

- Display data for both years on the same chart using different colored lines.
- Add markers at monthly data points.
- Include gridlines and month labels.
- Peaks and dips occurring in the same months across both years indicate seasonal trends.

(d) Explain how a diverging color scale could be used if defect rates were instead expressed as a deviation from a target rate (some suppliers above target, some below).
Answer:
A diverging color scale uses a neutral color for the target value and contrasting colors for values above and below the target.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Losses
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Wins
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Measure Values
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Team
Columns
Rows
SUM(Wins)
Highest number of wins among all teams.
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Wins
120
110
100
90
80
70
60
50
40
30
20
10
0
For a stacked bar chart use:
Measure
Dimension
optional Dimension on
Choose for me
Data Source
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